CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

VI Semester of BSc (MI) Examination April-May 2018

Course code & Name: MI603 Basic Agricultural Microbiology

Date: 04/05/2018 Day: Friday Time: 1:30 pm To 1:50 pm Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions. 10

1. Which of the following is free living nitrogen fixer?

(i) *Azotobacter* (ii) *Bacillus megaterium*

(iii) *Nostoc* (iv) All of these

2. Among the following biogeochemical cycle which one does not have a loss due to respiration?

(i) Sulphur (ii) Nitrogen

(iii) Phosphorus (iv) All of these

3. Which molecule plays an important role in the regulation of nitrogenase activity during day and night time in *R. rubrum*?

(i) ADP-ribosyl group (ii) cyclic AMP

(iii) ADP (iv) FAD

4. The advantage to plants during mycorrhizal association is ________.

(i) food (ii) N_2 fixation

(iii) Increase the mineral absorption and disease protection (iv) all of these

5. Generally, the R: S ratio is higher for ___________.

(i) fungi (ii) algae

(iii) bacteria (iv) protozoa

6. ______fungal bio-control is/are well known for controlling plant-pathogenic fungi.

(i) *Trichoderma* (ii) Beauveria

(iii) *B. thuringiensis* (iv) (i) and (ii) both

7. The viable cells count in the carrier based bacterial inoculants should be maintained up to ______as per ISI specifications.

(i) 10^9/gram (ii) 10^5/gram

(iii) 10^6 //gram (iv) 10^{20}/gram

8. ______ is an example of indirect antagonism.

(i)Antibiosis (ii) Lytic enzymes

(iii) Induction of host resistance (iv) Hyperparasitism

9. Azzola is used as biofertilizer as it has_________.

(i) Rhizobium (ii) cyanobacteria

(iii) mycorrhiza (iv) large quantity of humus

10. During the N_2 fixation process, which gene product is recognizes the flavonoids secreted by plant cell?

(i) nod A (ii) nod B

(iii) nod C (iv) nod D

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VI Semester of BSc (MI) Examination April-May 2018

Course code & Name: MI603 Basic Agricultural Microbiology

Date: 04/05/2018 Day: Friday Time: 1:51 pm To 3:30 pm Maximum Marks: 25

Instructions:

1. **Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

	SECTION -I	
Q - II	**Answer the following questions as directed**	**10**
(a)	What is bio control agent? Explain the direct and indirect antagonism mechanisms leading to biological control of plant pathogens.	**04**
(b)	Explain the nitrogenase oxidation protection mechanisms in free living bacteria, cyanobacteria and symbiotic microorganisms.	**04**
	or	
(b)	Describe the different types of mycorrhizal association and its importance in agriculture.	
(C)	Define: Saprophytism and Siderophores	**02**
	SECTION – II	
Q-III	**Answer the following questions as directed (any five)**	**15**
(a)	Write a note on “Carrier used in biofertilizers”.	
(b)	Explain in brief about biological control of insects by bacterial and fungal insecticides.	
(c)	What is biogeochemical cycle? Explain phosphorus and sulphur cycle.	
(d)	What biofertilizer? Discuss different criteria of strain selection used in biofertilizer production.	
(e)	Compare chemical pesticides Vs. bio pesticides	
(f)	Write a note on “microorganisms associated with organic matter decomposition”	